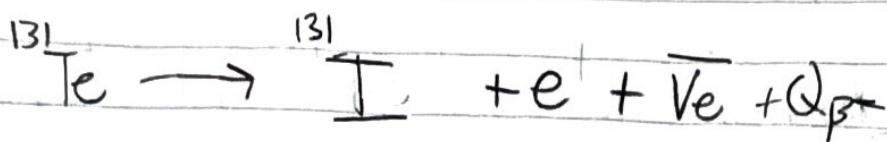


Parker
Lewis

1.)

$$A(t) = -\frac{dN}{dt} = \lambda N(t)$$



$$N_2(t) = \frac{\lambda_1}{\lambda_2 - \lambda_1} N_1^0 (e^{-\lambda_1 t} - e^{-\lambda_2 t}) + N_2^0 e^{-\lambda_2 t}$$

$$\frac{dN_2}{dt} = \frac{\lambda_1 N_1^0}{\lambda_2 - \lambda_1} \left[-\lambda_1 e^{-\lambda_1 t} + \lambda_2 e^{-\lambda_2 t} \right]$$

$$- \lambda_2 N_2^0 e^{-\lambda_2 t}$$

$$\frac{dN_2}{dt} = -\frac{\lambda_1^2 N_1^0}{\lambda_2 - \lambda_1} e^{-\lambda_1 t} + e^{-\lambda_2 t} \left[\frac{\lambda_2 \lambda_1 N_1^0 - \lambda_2 N_2^0}{\lambda_2 - \lambda_1} \right]$$

Use assumption at $t=0$, $N_2^0 = 0$
from problem.

$$N_2(t) = \frac{\lambda_1}{\lambda_2 - \lambda_1} N_1^0 (e^{-\lambda_1 t} - e^{-\lambda_2 t})$$

$$\frac{dN_2(t)}{dt} = \frac{\lambda_1}{\lambda_2 - \lambda_1} N_1^0 [-\lambda_1 e^{-\lambda_1 t} + \lambda_2 e^{-\lambda_2 t}]$$

Maximum is at $\frac{dN_2(t)}{dt} = 0$

$$0 = \lambda_2 e^{-\lambda_2 t} - \lambda_1 e^{-\lambda_1 t}$$

$$\lambda_1 e^{-\lambda_1 t} = \lambda_2 e^{-\lambda_2 t}$$

$$\frac{e^{-\lambda_1 t}}{e^{-\lambda_2 t}} = \frac{\lambda_2}{\lambda_1}$$

$$\ln\left(\frac{e^{-\lambda_1 t}}{e^{-\lambda_2 t}}\right) = \ln\left(\frac{\lambda_2}{\lambda_1}\right)$$

$$\ln(e^{-\lambda_1 t}) - \ln(e^{-\lambda_2 t}) = \ln\left(\frac{\lambda_2}{\lambda_1}\right)$$

$$- \lambda_1 t + \lambda_2 t = \ln\left(\frac{\lambda_2}{\lambda_1}\right)$$

6.136

$$t_{\max} = \frac{1}{(\lambda_2 - \lambda_1)} \ln\left(\frac{\lambda_2}{\lambda_1}\right)$$

* from nndc *

$$(t_{1/2})_{^{131}\text{Te}} = 26 \text{ min} \rightarrow 1500 \text{ s}$$

$$(t_{1/2})_{^{131}\text{I}} = 8.0247 \text{ days} \rightarrow 693334.08 \text{ s}$$

$$t_{1/2} = \frac{\ln(2)}{\lambda}$$

λ_1 λ_2

$$\lambda_1 = \frac{\ln(2)}{1500 \text{ s}} = 4.62 \times 10^{-4} \text{ s}^{-1}$$

$$\lambda_2 = \frac{\ln(2)}{693334.08 \text{ s}} = 9.997 \times 10^{-7} \text{ s}^{-1}$$

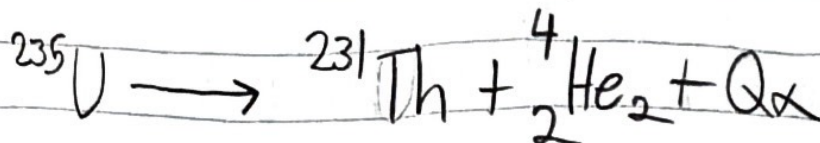
$$t_{\max} = 13310.19 \text{ s}$$

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^{235}U case:

HW #3

2.)



$$(Q_\alpha)_{\text{ndc}} = 4.6781 \text{ MeV}$$

$$W(\alpha) = 10^{-2}, \quad A_\alpha = 4, \quad A_d = 231$$

$$Z_\alpha = 2, \quad Z_d = 90$$

$$f = \frac{\sqrt{2(V_0 + Q_\alpha)}}{M} \frac{1}{2r_0(A_\alpha^{1/3} + A_d^{1/3})}$$

$$M = \frac{M_\alpha M_d}{M_\alpha + M_d}$$

$$(M_\alpha)_{\text{ndc}} = 4.001603 \text{ amu}, \quad V_0 = \frac{Z_\alpha Z_d e^2}{4\pi\epsilon_0 R}$$

$$(M_d)_{\text{ndc}} = 231.0363 \text{ amu}, \quad V_0 \approx 28 \text{ MeV}$$

$$M = 3.9344 \text{ amu}, \quad R = 9.27 \text{ fm}$$

important constants:

$$e^2 = 1.44 \text{ MeV}\cdot\text{fm}, \quad hc = 197.327 \text{ MeV}\cdot\text{fm}$$

$$r_0 = 1.2 \text{ fm}, \quad 1u = 931.494 \text{ MeV}/c^2$$

$$T = \exp\left(-2\pi \frac{e^2}{\hbar} Z_\alpha Z_d \sqrt{2Q_\alpha}\right)$$

$$r_0(A_\alpha^{1/3} + A_d^{1/3})$$

$$c = 3 \times 10^8 \text{ m/s}$$

$$f = \frac{c}{\sqrt{\frac{2(V_0 + Q_\alpha)}{\mu}}} \cdot \frac{1}{2R}$$

$$f = \frac{3 \times 10^8 \text{ m/s}}{\sqrt{\frac{2(28 + 4.678) \text{ MeV}}{3.9344 \left(\frac{931.4 \text{ MeV}}{c^2} \right)}}} \cdot \frac{1}{2(9.27 \times 10^{-15} \text{ m})}$$

$$f \approx 2.16 \times 10^{21} \text{ s}^{-1}$$

$$I = \exp \left(- \frac{2\pi e^2 Z_\alpha Z_d}{\hbar c} \sqrt{\frac{\mu c^2}{2Q_\alpha}} \left(1 - \frac{4}{\pi} \sqrt{\frac{6(A_\alpha^{1/3} + A_d^{1/3})}{2dZ_\alpha e^2}} \right) \right)$$

$$I = \exp \left(- \frac{2\pi(1.44 \text{ MeV-fm})}{(197.327 \text{ MeV-fm})} \frac{(2)(90)}{\sqrt{\frac{3.9344 \left(\frac{931.4 \text{ MeV}}{c^2} \right)}{2(4.678 \text{ MeV})}}} \right) \quad \times$$

$$\left(1 - \frac{4}{\pi} \sqrt{\frac{9.27 \text{ fm}}{\left(\frac{2(90)}{4.678 \text{ MeV}} \right) 1.44 \text{ MeV-fm}}} \right)$$

$$I = 1.012 \times 10^{-34}$$

$$t_{1/2} = \frac{\ln(2)}{\lambda}$$

$$\lambda_{^{235}\text{U}} = \left(1.012 \times 10^{-34}\right) \left(2.16 \times 10^{21} \text{ s}^{-1}\right) \left(10^{-2}\right)$$

$$\lambda_{^{235}\text{U}} = 2.189 \times 10^{-15} \text{ s}^{-1}$$

$$\begin{aligned} (t_{1/2})_{^{235}\text{U}} &\approx \frac{\ln(2)}{(2.189 \times 10^{-15} \text{ s}^{-1})} \\ &\approx 3.167 \times 10^{14} \text{ s} \end{aligned}$$

$$\begin{aligned} (t_{1/2})_{^{235}\text{U}} &\approx 3.167 \times 10^{14} \text{ s} \left(\frac{1 \text{ min}}{60 \text{ s}}\right) \left(\frac{1 \text{ hr}}{60 \text{ min}}\right) \\ &\quad \left(\frac{1 \text{ day}}{24 \text{ hr}}\right) \left(\frac{1 \text{ year}}{365 \text{ day}}\right) \end{aligned}$$

$$(t_{1/2})_{^{235}\text{U}} = 1.0042 \times 10^7 \text{ yrs}$$

$$(t_{1/2})_{^{235}\text{U}_{\text{nuc}}} = 7.04 \times 10^8 \text{ yrs}$$

from my Root program:

- 1.) U-235, $t_{1/2} \cong 1 \times 10^7 \text{ yrs}$
- 2.) Pa-231, $t_{1/2} \cong 1.6 \times 10^3 \text{ yrs}$
- 3.) Ac-227, $t_{1/2} \cong 7.86 \times 10^2 \text{ yrs}$
- 4.) Fr-223, $t_{1/2} \cong 6.168 \times 10^{-2} \text{ yrs}$
- 5.) At-219, $t_{1/2} \cong 1.07 \times 10^{-6} \text{ yrs}$

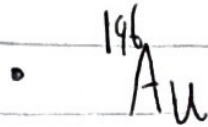
from nndc:

- 1.) U-235, $t_{1/2} = 7.04 \times 10^8 \text{ yrs}$
- 2.) Pa-231, $t_{1/2} = 3.274 \times 10^4 \text{ yrs}$
- 3.) Ac-227, $t_{1/2} = 21.7725 \text{ yrs}$
- 4.) Fr-223, $t_{1/2} = 21.99 \text{ min}$
- 5.) At-219, $t_{1/2} = 56 \text{ sec}$

Hw #3 Nuclear #1

Parker
Lewis

3.)



• Q-equations: $Z \rightarrow Z+1$

$M(X) - Au$

$\beta^- \rightarrow [X - Y] c^2, n \rightarrow p + e^- + \bar{\nu}_e$

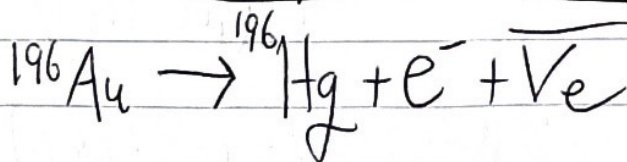
$\beta^+ \rightarrow [X - Y - 2m_0] c^2, p \rightarrow n + e^+ + \nu_e$
 $Z \rightarrow Z-1$

$EC \rightarrow [X - Y] c^2$

$Z \rightarrow Z-1, p + e^- \rightarrow n + \nu_e$

$2m_0 c^2 = 1.02 \text{ MeV}$

β^- Q-value:



$\Delta(^{196}\text{Au}) = -31138.7 \text{ keV}$

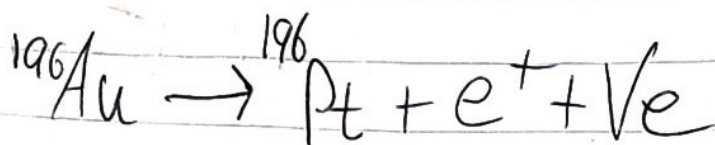
$\Delta(^{196}\text{Hg}) = -31825.9 \text{ keV}$

$$Q_{\beta^-} = [-31138.7 - (-31825.9)] \text{ keV}$$

$$Q_{\beta^-} = 687.2 \text{ keV} \left(\frac{1 \text{ MeV}}{1000 \text{ keV}} \right) = 0.6872 \text{ MeV}$$

from nndc $Q_{\beta^-} = 0.6872263 \text{ MeV}$

β^+ Q-value:

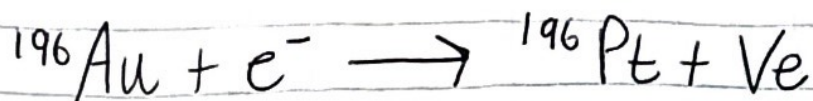


$$\Delta(^{196}\text{Pt}) = -32644.5 \left(\frac{1 \text{ MeV}}{1000 \text{ keV}} \right) = -32.6445 \text{ MeV}$$

$$Q_{\beta^+} = [-31138.7 - (-32.6445)] \text{ MeV}$$

$$Q_{\beta^+} = 0.4838 \text{ MeV}, Q_{\beta^+ \text{ (nndc)}} = 0.4838225 \text{ MeV}$$

EC - Q-value :



$$\Delta(^{196}\text{Au}) = -31.1387 \text{ MeV}$$

$$\Delta(^{196}\text{Pt}) = -32.6445 \text{ MeV}$$

$$Q_{\text{EC}} = [-31.1387 - (-32.6445)] \text{ MeV}$$

$$Q_{\text{EC}} = 1.5058 \text{ MeV}$$

$$Q_{\text{EC}}^{\text{(nndc)}} = 1.5018204 \text{ MeV}$$

Hw #3

4.)

a.) ${}^2\text{H}\left(\frac{1}{2}^{+}\right) \rightarrow {}^3\text{He}\left(\frac{1}{2}^{+}\right) + e^{-} + \bar{\nu}_e$

$$\Delta J = \frac{1}{2} - \frac{1}{2} = 0$$

$$\Delta \pi = \text{no}$$

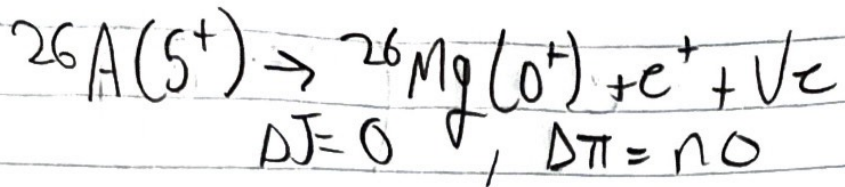
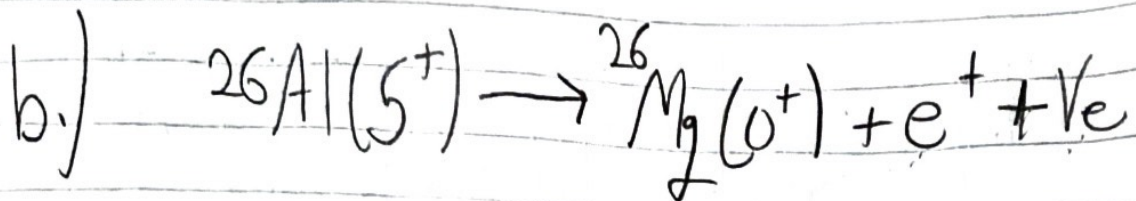
$$\pi = (-1)^L, \pi = + \text{ (both cases)}$$

$$J = L \pm 1/2$$

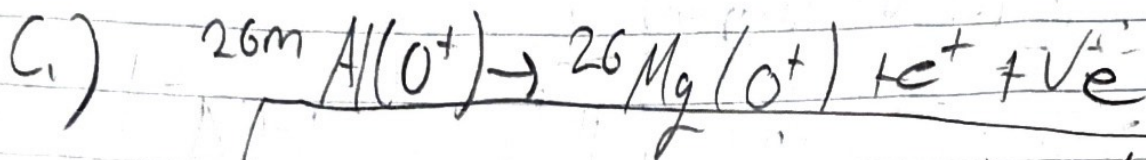
- $0^{+} \rightarrow 0^{+}$ is only superallowed case.

$\Delta L = 0$

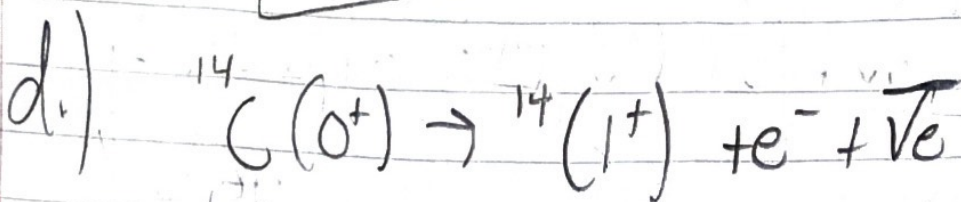
- $\Delta J = 0$, $\Delta \pi = \text{no}$,
it's allowed $\frac{1}{2}$, mixed state (Fermi's or GT)



$\Delta J = 5$, $\Delta \pi = \text{no}$
 4th forbidden



$0^+ \rightarrow 0^+$ it's superallowed
 and only fermi transition



$\Delta J = 1$, $\Delta \pi = \text{no}$
 Gamow-teller because $0^+ \rightarrow 1^+$
 allowed

ΔJ

Parker
Lavis

Hw #3

5.)

Part A.)

1.) Discuss input parameters

2.) At info missing or input values you don't need.

1.) Parent Information:

- Parent nucleus, Decay mode, E level (particular energy, transition), ΔE energy uncertainty, Q-value, $T_{1/2}$ and uncertainties for $T_{1/2}$ and Q-value.

Daughter Information:

E level, energy at specific quantum state, transition, ΔE energy uncertainty, $T_{1/2}$ is branching ratio, DTI is uniqueness, selection rule type.

2.) Only thing we don't need is uncertainties to get calculated f_t -values.

Part b.)

1.) Reproduce f_t values in paper format inputs.

2.) Comment on results, $\max \log(f_t)$ comment in terms of selection rules.

1.)

Reproduce $\log(f_t) = 4.25$:

Decay-Mod: β^-

$$Q_{\beta^-} = 10300 \text{ keV}$$

Parent $\rightarrow$ ^{54}Ca

$$T_{1/2} = 86 \text{ ms (unit)}$$

$$E_{\text{level (parent)}} = 0 \text{ keV}$$

$$E_{\text{level (daughter)}} = 247 \text{ keV}$$

$$\text{BR} = 97\%$$

(TI %)

Uniqueness $\rightarrow$ None

$$[\log(ft)]_{\text{nuc}} = 4.26$$

$$[\log(ft)]_{\text{paper}} = 4.25$$

$$\log(ft)_{\text{nuc}} \approx \log(ft)_{\text{paper}}$$

Reproduce 74.8 :

Decay Mode $\rightarrow \beta^-$
 $Q_{\beta^-} = 10300 \text{ keV}$
Parent $\rightarrow {}^{54}\text{Ca}$
 $E_{\text{level (parent)}} = 0 \text{ keV}$

$E_{\text{level (daughter)}} = [247, 110, 0]$
keV

BR
(TI %) = 32

$\log(ft) = [4.74, 4.77, 4.79]$

- Each case will close relate to paper $\log ft$ values

2.)

Selection Rules:

$(\log ft) \approx 4.26 \text{ } ^1_3 4.74, 4.77, 4.79$

all correspond to allowed T₁PR

Hw #3

Parker
Lewis

6.)

• γ -decay types

• $\frac{3}{2}^+$ state at $E_x = 4 \text{ MeV}$ in ^{19}Ne

$\frac{1}{2}^+$ GS.

• $|J_i - J_f| \leq l \leq |J_i + J_f|$

$\Delta\pi \rightarrow$ change

• $J_i - J_f = \frac{3}{2} - \frac{1}{2} = 1$

$J_i + J_f = \frac{3}{2} + \frac{1}{2} = 2$

• $1 \leq l \leq 2$
 $\Delta\pi = \text{no}$

Multipolarity types possible $\rightarrow$

• $M1 \quad 3 \quad E2$

$$A=19, \quad f_s = 10^{-15} \text{ s}$$

$$\underline{M1:} \quad 3.1 \times 10^{13} E_\gamma^3 = \lambda (s^{-1})$$

$$\underline{E2:} \quad 7.4 \times 10^7 A^{4/3} E_\gamma^5 = \lambda (s^{-1})$$

$$\begin{aligned} \lambda_{M1} (s^{-1}) &= 3.1 \times 10^{13} (4 \text{ MeV})^3 \\ &= 1.984 \times 10^{15} \end{aligned}$$

$$\begin{aligned} \lambda_{E2} (s^{-1}) &= 7.4 \times 10^7 (19)^{4/3} (4 \text{ MeV})^5 \quad 176 \text{ fs} \\ &= 3.84 \times 10^{12} \end{aligned}$$

$$(t_{1/2})_{M1} = \frac{\ln(2)}{(1.984 \times 10^{15} s^{-1})} = 3.49 \times 10^{-16} \text{ s}$$

$$(t_{1/2})_{E2} = \frac{\ln(2)}{(3.84 \times 10^{12} s^{-1})} = 1.805 \times 10^{-13} \text{ s} \quad (BR=0.8 \text{ for } G.S.)$$

$$(t_{1/2})_{nucle} \leq 35 \text{ fs} / 0.8 = 43.75 \text{ fs}$$

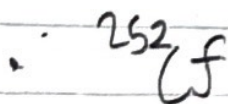
more closely aligned to M1

HW #3

- 7.) • Using shell model "magic number interpretation" One split of each will be most likely fission fragment splits to both parents.

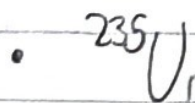
• This principle will be used:

"Asymmetric fission occurs because fission fragments are close to or at magic numbers".



$$Z = 98$$

$$N = 154$$



$$Z = 92$$

$$N = 143$$

• General Rule $Z \geq 98$ symmetric fission dominates both cases will mostly have asymmetric fission.
(f-252 will start to have symm contributions.)

Splits be stable:

asymmetric splits

1-heavy
1-light

- A, Z, N must be conserved

Cf - 252

252
Cf

$Z = 98$
 $N = 154$

try out

$Z = 50$

$N = 82$

$A = 132$

$^{132}_{50}\text{Sn}$

$A = 120$

$Z = 48$

$N = 72$

$^{120}_{48}\text{Cd}$

- One is heavy ✓
other is light ✓

$e^2 = 1.44 \text{ MeV-fm}$

- One fragment is doubly magic ✓

- A, Z, N is conserved ✓

- This configuration split works and is highly probable ✓

$$TKE = \frac{(50)(48)(1.44 \text{ MeV-fm})}{1.8(132^{1/3} + 120^{1/3}) \text{ fm}} = 190.54 \text{ MeV}$$

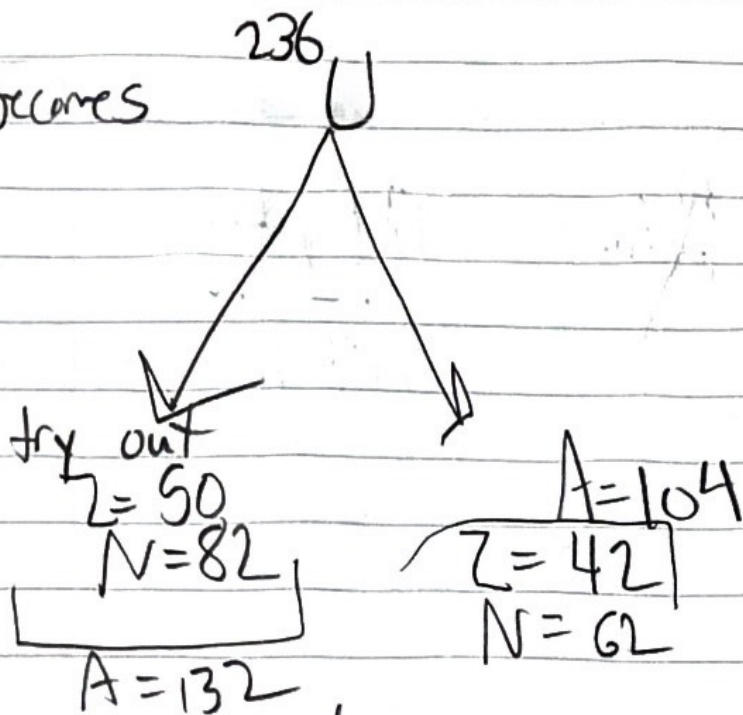
U-236 :

$Z=92$

$N=144$

• ^{235}U is stable so far
to fission add neutron " n "

• Parent becomes ^{236}U



✓ One is heavier while other is lighter.

✓ One fragment has the magic split.

✓ A, Z, N is conserved.

$$\text{TKE} = \frac{(50)(42)(1.44 \text{ MeV-fm})}{1.8 \left(132^{1/3} + 104^{1/3} \right)} = 171.53 \text{ MeV}$$

TKE of $^{252}\text{Cf} \rightarrow ^{235}\text{U}$

$$\text{Ratio} = \frac{(\text{TKE})_{^{252}\text{Cf}}}{(\text{TKE})_{^{235}\text{U}}} = \frac{190.54}{171.53} = 1.11$$

8.) Ungrading: Improvement

from Hw-2 based on putting more time and days into this one, more collaboration with classmates, only finish Ch 7-9 reading that I usually do for Hw. I think I'm in range 85-95.